

Statistical Inference for Linear Stochastic Approximation with Richardson-Romberg Extrapolation

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Abstract

Your abstract.

1. Introduction

Stochastic approximation (SA) algorithms are a cornerstone of modern computational statistics, optimization, and reinforcement learning. Introduced by Robbins and Monro (1951), these iterative procedures provide a principled way to find roots of equations or optimize objectives when only noisy observations are available. A particularly important subclass is the *linear stochastic approximation* (LSA) algorithm, which arises naturally in temporal-difference (TD) learning, policy evaluation, and stochastic gradient descent for linear models.

In this work, we study the LSA recursion with a *constant step size* $\alpha > 0$:

$$\theta_k^{(\alpha)} = \theta_{k-1}^{(\alpha)} - \alpha \{A(Z_k)\theta_{k-1}^{(\alpha)} - b(Z_k)\}, \quad k \geq 1,$$

where $\{Z_k\}_{k \in \mathbb{N}}$ is a time-homogeneous Markov chain on a measurable space $(Z, \mathcal{Z})$ with transition kernel Q and unique invariant distribution π . The mappings $A : Z \rightarrow \mathbb{R}^{d \times d}$ and $b : Z \rightarrow \mathbb{R}^d$ are measurable functions satisfying $\bar{A} := \int_Z A(z) d\pi(z)$ and $\bar{b} := \int_Z b(z) d\pi(z)$. Assuming that $-\bar{A}$ is a *Hurwitz matrix* (all eigenvalues have strictly negative real parts), the target parameter $\theta^* = \bar{A}^{-1}\bar{b}$ is uniquely defined.

The *Polyak–Ruppert averaging* procedure (Polyak, 1990; Ruppert, 1988) provides an effective variance reduction technique. Given a burn-in period $n_0 \geq 0$, the averaged iterate is defined as

$$\bar{\theta}_n^{(\alpha)} = \frac{1}{n - n_0} \sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}.$$

1.1. Bias of constant step-size iterates

The use of a constant step size $\alpha > 0$ offers several practical advantages: it enables geometrically fast forgetting of the initial condition (Dieuleveut, Durmus, and Bach, 2020) and simplifies hyperparameter tuning compared to diminishing step-size schedules. However, unlike the classical regime $\alpha_k \rightarrow 0$ with $\sum \alpha_k = \infty$ and $\sum \alpha_k^2 < \infty$, a constant step size produces iterates that converge only *in distribution* to a stationary measure Π_α , rather than almost surely to θ^* . The stationary expectation $\mathbb{E}[\theta_\infty^{(\alpha)}]$ is generally *biased* with respect to θ^* , and this bias cannot be eliminated by Polyak–Ruppert averaging alone.

As shown in Levin, Naumov, and Samsonov (2025) and Huo, Chen, and Xie (2023), the asymptotic bias admits a power-series expansion in α :

$$\lim_{n \rightarrow \infty} \mathbb{E}[\theta_n^{(\alpha)}] = \theta^* + \alpha \Delta + O(\alpha^2),$$

where $\Delta = \bar{A}^{-1} \sum_{k=1}^{\infty} \mathbb{E}[\{Q^k \tilde{A}(Z_{\infty})\} \epsilon(Z_{\infty})]$ depends on the correlation structure of the Markov chain, and $\tilde{A}(z) = A(z) - \bar{A}$ is the centered matrix-valued function.

1.2. Richardson–Romberg extrapolation

To eliminate the leading $O(\alpha)$ bias term, we employ the *Richardson–Romberg* (RR) *extrapolation* procedure. Two LSA sequences are run *on the same Markov chain trajectory* $\{Z_k\}$ with step sizes α and 2α , and the RR iterate is formed as

$$\bar{\theta}_n^{(\alpha, \text{RR})} = 2\bar{\theta}_n^{(\alpha)} - \bar{\theta}_n^{(2\alpha)}.$$

Since both sequences share the same noise realization, the leading bias term $\alpha\Delta$ cancels, leaving a residual bias of order $O(\alpha^{3/2})$ or higher (Levin et al., 2025).

More generally, one can consider the multi-level extrapolation with M step sizes $\mathcal{A} = \{\alpha_1, \dots, \alpha_M\}$ and coefficients $\{h_m\}$ determined by the Vandermonde system (Huo et al., 2023):

$$\sum_{m=1}^M h_m = 1, \quad \sum_{m=1}^M h_m \alpha_m^l = 0, \quad l = 1, \dots, M-1,$$

which reduces the bias from $O(\alpha)$ to $O(\alpha^M)$.

1.3. Problem statement and goals

The high-order moment bounds for the RR iterate $\bar{\theta}_n^{(\alpha, \text{RR})}$ established in Levin et al. (2025) show that the leading error term scales as $\sqrt{\text{Tr } \Sigma_{\epsilon}^{(M)}} \cdot n^{-1/2}$, matching the minimax optimal rate. Berry–Esseen type bounds and bootstrap inference procedures for the *standard* Polyak–Ruppert average $\bar{\theta}_n$ (without extrapolation) under Markovian noise have been obtained in Samsonov, Sheshukova, Moulines, and Naumov (2025).

However, the *distributional approximation* — specifically, the central limit theorem and Berry–Esseen bounds — for the *averaged Richardson–Romberg iterates* $\bar{\theta}_n^{(\alpha, \text{RR})}$ remains an open problem. The main goal of this work is to fill this gap by establishing:

1. A *central limit theorem* for $\sqrt{n}(\bar{\theta}_n^{(\alpha, \text{RR})} - \theta^*)$, identifying the limiting covariance matrix.
2. *Non-asymptotic Berry–Esseen bounds* for the rate of convergence to the Gaussian limit.
3. A practical *inference procedure* (confidence intervals) based on the RR-extrapolated iterates.

1.4. Setting and assumptions

We now formalize the setting and state the assumptions that will be used throughout this work.

Let $\{Z_k\}_{k \in \mathbb{N}}$ be a Markov chain on a complete separable metric space $(Z, \mathcal{Z})$ with transition kernel Q .

Assumption 1 (Uniform geometric ergodicity). The kernel Q admits a unique invariant distribution π and is *uniformly geometrically ergodic*: there exists $t_{\text{mix}} \in \mathbb{N}^*$ such that for all $k \in \mathbb{N}^*$,

$$\Delta(Q^k) := \sup_{z, z' \in Z} \frac{1}{2} \|Q^k(z, \cdot) - Q^k(z', \cdot)\|_{\text{TV}} \leq (1/4)^{\lfloor k/t_{\text{mix}} \rfloor}.$$

Equivalently, there exist constants $\zeta > 0$ and $\rho \in (0, 1)$ such that $\sup_z \|Q^k(z, \cdot) - \pi\|_{\text{TV}} \leq \zeta \rho^k$ for all $k \geq 1$.

Assumption 2 (Hurwitz condition and boundedness). The matrix $-\bar{A}$ is Hurwitz, i.e., all eigenvalues of $\bar{A}$ have strictly positive real parts. Moreover,

$$C_A := \sup_{z \in Z} \|A(z)\| \vee \sup_{z \in Z} \|\tilde{A}(z)\| < \infty,$$

where $\tilde{A}(z) = A(z) - \bar{A}$.

Assumption 3 (Noise regularity). The noise function $\epsilon(z) = \tilde{A}(z)\theta^* - \tilde{b}(z)$, where $\tilde{b}(z) = b(z) - \bar{b}$, satisfies

$$\|\epsilon\|_\infty := \sup_{z \in Z} \|\epsilon(z)\| < +\infty.$$

Under Assumptions 1–3, the error $\theta_k^{(\alpha)} - \theta^*$ satisfies the recursion

$$\theta_k^{(\alpha)} - \theta^* = (I - \alpha A(Z_k))(\theta_{k-1}^{(\alpha)} - \theta^*) - \alpha \epsilon(Z_k).$$

1.5. Key quantities

The *Markovian noise covariance matrix* captures both the marginal variance and the temporal correlations of the noise:

$$\Sigma_\epsilon^{(M)} = \mathbb{E}_\pi [\epsilon(Z_0)\epsilon(Z_0)^\top] + 2 \sum_{\ell=1}^{\infty} \mathbb{E}_\pi [\epsilon(Z_0)\epsilon(Z_\ell)^\top].$$

This matrix is the limiting covariance in the Markov chain CLT for the partial sums $n^{-1/2} \sum_{t=0}^{n-1} \epsilon(Z_t)$ (cf. Douc et al., 2018, Theorem 21.2.10).

The *asymptotically optimal covariance matrix* is given by

$$\Sigma_\infty = \bar{A}^{-1} \Sigma_\epsilon^{(M)} (\bar{A}^{-1})^\top.$$

This covariance is optimal in the Hájek–Le Cam sense and represents the best achievable asymptotic variance for estimating θ^* via averaged LSA.

The *Lyapunov equation* plays a central role in the contraction analysis. For any $P = P^\top \succ 0$, there exists a unique $Q = Q^\top \succ 0$ satisfying $\bar{A}^\top Q + Q \bar{A} = P$. Defining $a = \lambda_{\min}(P)/(2 \|Q\|)$ and $\kappa_Q = \lambda_{\max}(Q)/\lambda_{\min}(Q)$, the key contraction property holds: for all $\alpha \in [0, \alpha_\infty]$,

$$\|I - \alpha \bar{A}\|_Q^2 \leq 1 - \alpha a.$$

Since the iterates $\theta_k^{(\alpha)}$ alone are generally not Markovian (due to the Markovian noise), we consider the *joint process* $(\theta_k^{(\alpha)}, Z_{k+1})$ with kernel

$$\bar{P}_\alpha f(\theta, z) = \int_Z Q(z, dz') f(F_{z'}(\theta), z'),$$

where $F_z(\theta) = (I - \alpha A(z))\theta + \alpha b(z)$. Under Assumptions 1–3, this joint chain admits a unique invariant distribution Π_α for sufficiently small $\alpha > 0$ (Levin et al., 2025).

2. Perturbation Expansion

2.1. LSA Error Decomposition

We consider the recursion

$$\theta_k = \theta_{k-1} - \alpha_k (A(Z_k)\theta_{k-1} - b(Z_k)), \quad \alpha_k = \alpha = \text{const.}$$

Define the transition products

$$\Gamma_{m:k} = \prod_{l=m}^k (I - \alpha A(Z_l)).$$

We also introduce

$$J_k^{(0)} = - \sum_{l=1}^k \alpha \Gamma_{l+1:k} \epsilon(Z_l), \quad H_k^{(0)} = - \sum_{l=1}^k \alpha \Gamma_{l+1:k} (A(Z_l) - \bar{A}) J_{l-1}^{(0)},$$

with initial conditions $J_0^{(0)} = H_0^{(0)} = 0$.

A standard decomposition yields

$$\sqrt{n}(\theta_n^{(\alpha)} - \theta^*) = W + D_1,$$

where

$$W = \frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l \epsilon(Z_l), \quad D_1 = \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \Gamma_{1:k} (\theta_0 - \theta^*) + \frac{1}{\sqrt{n}} \sum_{k=1}^{n-1} J_k^{(0)},$$

and

$$Q_l = \alpha \sum_{k=l}^{n-1} (I - \alpha \bar{A})^{n-k-1}.$$

2.2. RR Combination and the $\tilde{J}_n^{(0,\alpha)}$ Term

Applying the decomposition of the previous subsection separately to step sizes α and 2α and forming the Richardson–Romberg combination, we obtain

$$\begin{aligned} \theta_n^{(\text{RR},\alpha)} - \theta^* &= \left[2\Gamma_{1:n}^{(\alpha)} - \Gamma_{1:n}^{(2\alpha)} \right] (\theta_0 - \theta^*) \\ &\quad + \left[2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)} \right] + \left[2J_n^{(1,\alpha)} - J_n^{(1,2\alpha)} \right] + \left[2H_n^{(1,\alpha)} - H_n^{(1,2\alpha)} \right]. \end{aligned}$$

The first bracket is the deterministic transient and is exponentially small. The remaining three brackets are stochastic and need separate analysis. In this subsection we focus on the zeroth-order RR difference, which we denote

$$\tilde{J}_n^{(0,\alpha)} = 2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)},$$

where, by definition,

$$J_n^{(0,\alpha)} = -\alpha \sum_{j=1}^n (I - \alpha \bar{A})^{n-j} \epsilon(Z_j).$$

Substituting and using the elementary identity $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$ with $X = I - \alpha \bar{A}$, $Y = I - 2\alpha \bar{A}$, $X - Y = \alpha \bar{A}$, we get

$$\begin{aligned}
\tilde{J}_n^{(0,\alpha)} &= -2\alpha \sum_{j=1}^n \left[(I - \alpha \bar{A})^{n-j} - (I - 2\alpha \bar{A})^{n-j} \right] \epsilon(Z_j) \\
&= -2\alpha^2 \bar{A} \sum_{j=1}^n \underbrace{\sum_{i=1}^{n-j} (I - \alpha \bar{A})^{i-1} (I - 2\alpha \bar{A})^{n-j-i}}_{=: H_j^{(n)}} \epsilon(Z_j) \\
&= -2\alpha^2 \bar{A} \sum_{j=1}^n H_j^{(n)} \epsilon(Z_j).
\end{aligned}$$

The extra factor $\alpha \bar{A}$ pulled out front is the source of the additional α -decay of the RR difference compared to a single LSA trajectory.

2.3. Norm Estimate for $H_j^{(n)}$

To bound $\|H_j^{(n)}\|$ we use the following standard result.

Lemma 1. Let $-\bar{A}$ be a Hurwitz matrix. Then for any $P = P^\top \succ 0$ there exists a unique $Q = Q^\top \succ 0$ satisfying

$$\bar{A}^\top Q + Q \bar{A} = P.$$

Moreover, letting

$$a = \frac{\lambda_{\min}(P)}{2\|Q\|}, \quad \alpha_\infty = \frac{\lambda_{\min}(P)}{2\kappa_Q \|\bar{A}\|_Q^2} \wedge \frac{\|Q\|}{\lambda_{\min}(P)},$$

with $\kappa_Q = \lambda_{\max}(Q)/\lambda_{\min}(Q)$, one has for all $\alpha \in [0, \alpha_\infty]$:

$$\alpha a \leq 1/2, \quad \|I - \alpha \bar{A}\|_Q^2 \leq 1 - \alpha a.$$

We now estimate $\|H_j^{(n)}\|$ by combining the triangle inequality, submultiplicativity, the Lyapunov contraction (applied at step sizes α and 2α), and the equivalence of $\|\cdot\|$ and $\|\cdot\|_Q$ which absorbs κ_Q into a generic constant C_A :

$$\begin{aligned}
\|H_j^{(n)}\| &\leq \sum_{i=1}^{n-j} \|I - \alpha \bar{A}\|^{i-1} \|I - 2\alpha \bar{A}\|^{n-j-i} && \text{(triangle + submultiplicativity)} \\
&\leq C_A \sum_{i=1}^{n-j} (1 - \alpha a)^{(i-1)/2} (1 - 2\alpha a)^{(n-j-i)/2} && \text{(Lyapunov contraction)} \\
&= C_A (1 - \alpha a)^{(n-j-1)/2} \sum_{k=0}^{n-j-1} \left(\frac{1 - 2\alpha a}{1 - \alpha a} \right)^{k/2} && \text{(reindex } k = n - j - i) \\
&\leq C_A (1 - \alpha a)^{(n-j-1)/2} \frac{1}{1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}}} && \text{(geometric series).}
\end{aligned}$$

It remains to bound the geometric rate $1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}}$ from below. Writing $\frac{1-2\alpha a}{1-\alpha a} = 1 - \alpha \frac{a}{1-\alpha a}$ and Taylor-expanding $\sqrt{1-x}$ at $x = \alpha \frac{a}{1-\alpha a}$ gives

$$\sqrt{\frac{1-2\alpha a}{1-\alpha a}} = 1 - \frac{\alpha a}{2(1-\alpha a)} + O((\alpha a)^2) = 1 - \frac{1}{2}\alpha a + O(\alpha^2),$$

where the second equality uses $\alpha a \leq 1/2$. Consequently,

$$1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}} = \frac{1}{2}\alpha a + O(\alpha^2), \quad \frac{1}{1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}}} = \frac{2}{\alpha a} + O(1).$$

Folding the $O(1)$ correction into a constant $\bar{C}_A$, we arrive at the final estimate

$$\|H_j^{(n)}\| \leq \bar{C}_A (1 - \alpha a)^{(n-j-1)/2} \frac{2}{\alpha a}.$$

The kernel decays geometrically in $n - j$ at rate $\sqrt{1 - \alpha a}$ but its summed weight is of order $1/(\alpha a)$. The next subsection shows how the prefactor α^2 in $\tilde{J}_n^{(0,\alpha)}$ exactly compensates this divergence.

2.4. L^2 Bound for the Zeroth-Order Term

The expression

$$\tilde{J}_n^{(0,\alpha)} = - \sum_{j=1}^n 2\alpha^2 \bar{A} H_j^{(n)} \epsilon(Z_j)$$

is a weighted sum of values of the centered noise function ϵ along the Markov chain. We control its tail via the following Markov concentration statement.

Lemma 2. Assume **UGE 1**. Let $\{g_i\}_{i=1}^n$ be a family of measurable functions $g_i : Z \rightarrow \mathbb{R}^d$ such that

$$c_i = \|g_i\|_\infty < \infty \quad \text{for all } i \geq 1, \quad \pi(g_i) = 0 \quad \text{for all } i \in \{1, \dots, n\}.$$

Then, for any initial distribution ξ on $(Z, \mathcal{Z})$, any $n \in \mathbb{N}$ and any $t \geq 0$,

$$\mathbb{P}_\xi \left(\left\| \sum_{i=1}^n g_i(Z_i) \right\| \geq t \right) \leq 2 \exp \left(- \frac{t^2}{2u_n^2} \right),$$

where

$$u_n = 8 \left(\sum_{i=1}^n c_i^2 \right)^{1/2} \sqrt{t_{\text{mix}}}.$$

Proof. The proof follows the lines of Durmus et al. (2025, Lemma 9).

We apply the lemma with $g_{j(z)} = -2\alpha^2 \bar{A} H_j^{(n)} \epsilon(z)$. Each g_j is centered under π since $\pi(\epsilon) = 0$, so the centering hypothesis holds. Plugging the bound on $\|H_j^{(n)}\|$ from the previous subsection and writing $\tilde{C}_A = \|A\| \bar{C}_A$ yields the per-summand bound

$$\|g_j\|_\infty \leq 2\alpha^2 \tilde{C}_A (1 - \alpha a)^{(n-j-1)/2} \frac{2}{\alpha a} \|\epsilon\|_\infty = 4\alpha \tilde{C}_A \|\epsilon\|_\infty (1 - \alpha a)^{(n-j-1)/2}.$$

Note the prefactor α^2 collapsing to α : the $1/(\alpha a)$ blow-up in $H_j^{(n)}$ is exactly absorbed by one factor of α . Squaring and summing over j , we get a geometric series whose closed form is bounded uniformly in n :

$$\begin{aligned} \sum_{j=1}^n \|g_j\|_\infty^2 &\leq 16\alpha^2 \tilde{C}_A^2 \|\epsilon\|_\infty^2 \sum_{j=1}^n (1 - \alpha a)^{n-j-1} \\ &\leq 16\alpha^2 \tilde{C}_A^2 \|\epsilon\|_\infty^2 \frac{1}{(1 - \alpha a)\alpha a} \leq \frac{C^2}{a} \alpha \|\epsilon\|_\infty^2, \end{aligned}$$

where in the last step we used $1 - \alpha a \geq 1/2$. Plugging this into the variance proxy u_n from the lemma absorbs the constants $64 \cdot 16$ and $\sqrt{t_{\text{mix}}}$ into a constant $\hat{C}_A$ depending on a , t_{mix} , and $\|\epsilon\|_\infty$:

$$u_n^2 \leq \hat{C}_A^2 \alpha.$$

To convert the sub-Gaussian tail into a moment bound, we use the following standard fact.

Lemma 3. Let X be an $\mathbb{R}^d$ -valued random variable satisfying

$$\mathbb{P}(\|X\| \geq t) \leq 2 \exp\left(-\frac{t^2}{2\sigma^2}\right) \quad \text{for all } t \geq 0,$$

for some $\sigma^2 > 0$. Then, for any $p \geq 2$, it holds that

$$\mathbb{E}[\|X\|^p] \leq 2 p^{p/2} \sigma^p.$$

Applying the lemma with $\sigma^2 = u_n^2 \leq \hat{C}_A^2 \alpha$ and $p = 2$ gives

$$\mathbb{E}^{1/2}[\|\tilde{J}_n^{(0,\alpha)}\|^2] \leq 2\sqrt{\alpha} \hat{C}_A, \quad \text{i.e.} \quad \frac{1}{\sqrt{\alpha}} \mathbb{E}^{1/2}[\|\tilde{J}_n^{(0,\alpha)}\|^2] \leq 2\hat{C}_A.$$

The zeroth-order RR difference is therefore $O(\sqrt{\alpha})$ in L^2 , uniformly in n — an order- $\sqrt{\alpha}$ gain over a single LSA trajectory, which is $O(1)$ in L^2 in the stationary regime.

2.5. Richardson–Romberg PR Weight Bounds

The Polyak–Ruppert version of the LSA expansion replaces the trajectory $\theta_k - \theta^*$ by its time average and rewrites the leading martingale contribution as a single sum

$$W = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l^{(\alpha)} \epsilon(Z_l), \quad Q_l^{(\alpha)} = \alpha \sum_{k=l}^{n-1} (I - \alpha \bar{A})^{n-k-1},$$

so that $Q_l^{(\alpha)}$ collects the deterministic-product weight with which the noise $\epsilon(Z_l)$ enters the PR average. (For constant step this is the specialization of the time-dependent kernel introduced in Samsonov et al. (2025).) Coupling two trajectories with steps α and 2α through the same noise realization $\{Z_k\}$ and forming the Richardson–Romberg combination,

$$Q_l^{\text{RR}} := 2Q_l^{(\alpha)} - Q_l^{(2\alpha)},$$

we obtain the leading martingale weight for $\sqrt{n}(|(\theta)_n^{(\text{RR},\alpha)} - \theta^*)|$. The Berry–Esseen analysis of the RR-averaged iterate hinges on two contraction estimates for Q_l^{RR} : comparison with the asymptotic weight $\bar{A}^{-1}$, and a bound on the successive difference $Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}$. The latter controls the Abel-summation term in the Poisson-equation remainder, which is the structural reason that constant-step RR achieves the same Berry–Esseen rate as diminishing-step PR (Samsonov et al. 2025).

Throughout this subsection write

$$B_\alpha := I - \alpha \bar{A}, \quad B_{2\alpha} := I - 2\alpha \bar{A}, \quad k := n - l.$$

We assume $\alpha, 2\alpha \in (0, \alpha_\infty]$, so that the Lyapunov contraction $\|B_\alpha^m\|_Q^2 \leq 1 - \alpha a$ and $\|B_{2\alpha}^m\|_Q^2 \leq 1 - 2\alpha a$ hold (and we use freely $1 - 2\alpha a \leq 1 - \alpha a$).

The geometric-series identity gives the closed form

$$Q_l^{(\alpha)} = \alpha \sum_{j=0}^{n-l-1} B_\alpha^j = \alpha (\alpha \bar{A})^{-1} (I - B_\alpha^{n-l}) = \bar{A}^{-1} (I - B_\alpha^{n-l}).$$

Two immediate consequences are

$$Q_l^{(\alpha)} - \bar{A}^{-1} = -\bar{A}^{-1} B_\alpha^{n-l}, \quad Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)} = -\alpha B_\alpha^{n-l-1}.$$

The first identity, applied to α and 2α , yields the basic RR identity

$$\mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1} = -\bar{A}^{-1} (2B_\alpha^k - B_{2\alpha}^k),$$

and the second yields the discrete-difference identity

$$\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}} = -2\alpha (B_\alpha^{k-1} - B_{2\alpha}^{k-1}).$$

Lemma 4. Let $\alpha, 2\alpha \in (0, \alpha_\infty]$, set $C_Q := \kappa_Q^{1/2} \|\bar{A}^{-1}\|$ and $\tilde{C}_A := \kappa_Q \|\bar{A}\|$, and write $k = n - l$.

(i) For every $1 \leq l \leq n - 1$,

$$\|\mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1}\| \leq 3C_Q(1 - \alpha a)^{k/2}.$$

(ii) For every $0 \leq l \leq n - 2$,

$$\|\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}}\| \leq 2\tilde{C}_A \alpha^2 (k - 1) (1 - \alpha a)^{(k-2)/2}.$$

Proof of (i). Take norms in the basic RR identity above. Using the equivalence $\|\cdot\| \leq \kappa_Q^{1/2} \|\cdot\|_Q$ and the Lyapunov contraction at both step sizes,

$$\|2B_\alpha^k - B_{2\alpha}^k\|_Q \leq 2\|B_\alpha^k\|_Q + \|B_{2\alpha}^k\|_Q \leq 2(1 - \alpha a)^{k/2} + (1 - 2\alpha a)^{k/2} \leq 3(1 - \alpha a)^{k/2}.$$

Submultiplicativity in $\|\cdot\|_Q$ followed by norm equivalence then gives

$$\|\mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1}\| \leq \kappa_Q^{1/2} \|\bar{A}^{-1}\| \cdot \|2B_\alpha^k - B_{2\alpha}^k\|_Q \leq 3C_Q(1 - \alpha a)^{k/2}.$$

Proof of (ii). Apply the elementary identity $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$ with $X = B_\alpha$, $Y = B_{2\alpha}$, $X - Y = \alpha\bar{A}$, and $m = k - 1$:

$$B_\alpha^{k-1} - B_{2\alpha}^{k-1} = \alpha\bar{A} \sum_{i=1}^{k-1} B_\alpha^{i-1} B_{2\alpha}^{k-1-i}.$$

Each summand obeys, by submultiplicativity in $\|\cdot\|_Q$ and Lyapunov contraction,

$$\begin{aligned} \|B_\alpha^{i-1} B_{2\alpha}^{k-1-i}\| &\leq \kappa_Q^{1/2} (1 - \alpha a)^{(i-1)/2} (1 - 2\alpha a)^{(k-1-i)/2} \\ &\leq \kappa_Q^{1/2} (1 - \alpha a)^{(k-2)/2}, \end{aligned}$$

where in the last step we used $1 - 2\alpha a \leq 1 - \alpha a$. Summing over $i \in \{1, \dots, k - 1\}$ and absorbing constants,

$$\|B_\alpha^{k-1} - B_{2\alpha}^{k-1}\| \leq \alpha \kappa_Q \|\bar{A}\| (k - 1) (1 - \alpha a)^{(k-2)/2}.$$

Inserting this bound into the discrete-difference identity finishes the proof. $\square$

Corollary 1. Under the assumptions of the previous lemma, uniformly in $n \geq 2$,

$$\sum_{l=1}^{n-1} \|\mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1}\|^2 \leq \frac{C_1}{\alpha a}, \quad \sum_{l=0}^{n-2} \|\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}}\| \leq \frac{C_2}{a^2},$$

with $C_1 = 9C_Q^2$ and $C_2 = 32\tilde{C}_A$.

Proof. For the first sum apply (i) and the geometric series $\sum_{k \geq 1} (1 - \alpha a)^k \leq \frac{1}{\alpha a}$. For the second, apply (ii) and

$$\sum_{k \geq 1} k (1 - \alpha a)^{(k-2)/2} \leq \frac{2}{1 - \alpha a} \sum_{m \geq 0} (m + 1) (1 - \alpha a)^{m/2} \leq \frac{2}{1 - \alpha a} \cdot \frac{1}{(1 - \sqrt{1 - \alpha a})^2} \leq \frac{16}{(\alpha a)^2},$$

where the last step uses $1 - \sqrt{1 - \alpha a} \geq \alpha \frac{a}{2}$ for $\alpha a \leq 1/2$. Multiplying by $2\tilde{C}_A \alpha^2$ gives the claim. $\square$

Remark 1. Comparison with the single-step case. The same identities for $Q_l^{(\alpha)}$ give $\|Q_l^{(\alpha)} - \bar{A}^{-1}\| \leq C_Q(1 - \alpha a)^{k/2}$ and $\|Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)}\| \leq \alpha \kappa_Q^{1/2} (1 - \alpha a)^{(k-1)/2}$. Part (i) of the lemma above is therefore the same order as the single-step bound: the leading $-\bar{A}^{-1}$ term of Q_{n-1}^{RR} is exactly $-\bar{A}^{-1}$ (since $Q_{n-1}^{\text{RR}} = 0$ identically), so no factor of α is gained on the boundary. The Richardson–Romberg cancellation is genuinely visible only in the successive difference of (ii): the discrete derivative gains an extra factor of α , becoming $O(\alpha^2)$ rather than $O(\alpha)$. This extra α feeds into the Poisson-equation Abel-summation term D_2^{RR} and is the structural place where constant-step RR matches the diminishing-step PR rate. The improvement on the variance side is recovered by squaring and summing in the corollary, which gives $\sum \|Q_l^{\text{RR}} - \bar{A}^{-1}\|^2 = O(\frac{1}{\alpha a})$ — exactly enough for the linear part of $\Sigma_n^{\text{RR}} - \Sigma_\infty$ to be $O(\frac{1}{n\alpha a})$.

3. Last Iterate Analysis

3.1. Centered Bound for the Shifted First-Order Perturbation

The purpose of this section is to isolate the last-iteration weighted term which appears in the analysis of $J_n^{(1,\alpha)}$, and to record a clean L_p bound for its centered part. The proof uses the deterministic-product perturbation expansion from Samsonov et al. (2025, Proposition 9), specialized to the constant-stepsize setting.

Let

$$B = I - \alpha \bar{A}.$$

Throughout this section we assume $0 < \alpha \leq \alpha_\infty$, so that

$$\|B^m\|_Q \leq (1 - \alpha a)^{m/2}, \quad m \geq 0.$$

All constants denoted by C may depend on C_A , κ_Q , and norm equivalence constants, but not on α , n , p , or t_{mix} .

The deterministic-product components are defined by

$$J_n^{(0,\alpha)} = B J_{n-1}^{(0,\alpha)} - \alpha \epsilon(Z_n), \quad J_0^{(0,\alpha)} = 0,$$

and

$$J_n^{(1,\alpha)} = B J_{n-1}^{(1,\alpha)} - \alpha \tilde{A}(Z_n) J_{n-1}^{(0,\alpha)}, \quad J_0^{(1,\alpha)} = 0.$$

Hence

$$J_n^{(0,\alpha)} = -\alpha \sum_{k=1}^n B^{n-k} \epsilon(Z_k),$$

and

$$\begin{aligned} J_n^{(1,\alpha)} &= -\alpha \sum_{j=1}^n B^{n-j} \tilde{A}(Z_j) J_{j-1}^{(0,\alpha)} \\ &= \alpha^2 \sum_{1 \leq k < j \leq n} B^{n-j} \tilde{A}(Z_j) B^{j-1-k} \epsilon(Z_k) \\ &= \alpha^2 \sum_{k=1}^{n-1} \sum_{r=1}^{n-k} B^{n-k-r} \tilde{A}(Z_{k+r}) B^{r-1} \epsilon(Z_k). \end{aligned}$$

The sign in the second line is positive because $J_{j-1}^{(0,\alpha)}$ already contains the minus sign.

The last display is for $J_n^{(1,\alpha)}$ itself. In the last-iteration estimate below we use one additional left factor B :

$$S_n = \sum_{t=0}^{n-1} B^{n-t} \tilde{A}(Z_{t+1}) J_t^{(0,\alpha)}.$$

Thus the corresponding shifted first-order contribution is

$$T_n^{(1,\alpha)} = -\alpha S_n.$$

Lemma 5. Assume the Markov chain is started from stationarity, that is, the law of Z_1 is π , and $\pi(\tilde{A}) = 0$. For every $p \geq 2$,

$$\|S_n - \mathbb{E}S_n\|_{L_p} \leq C \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right).$$

Consequently,

$$\|T_n^{(1,\alpha)} - \mathbb{E}T_n^{(1,\alpha)}\|_{L_p} \leq C\alpha \|\epsilon\|_\infty \left(p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right).$$

Proof. Since $J_0^{(0,\alpha)} = 0$, the term $t = 0$ in S_n vanishes. Substituting the explicit formula for $J_t^{(0,\alpha)}$ gives

$$\begin{aligned} S_n &= -\alpha \sum_{t=1}^{n-1} \sum_{k=1}^t B^{n-t} \tilde{A}(Z_{t+1}) B^{t-k} \epsilon(Z_k) \\ &= -\alpha \sum_{k=1}^{n-1} H_{k+1}^{(w)} \epsilon(Z_k), \end{aligned}$$

where, after the change of summation index $l = t - k + 1$,

$$H_{k+1}^{(w)} = \sum_{l=1}^{n-k} B^{n-k-l+1} \tilde{A}(Z_{k+l}) B^{l-1}.$$

The kernel $H_{k+1}^{(w)}$ acts on the past noise $\epsilon(Z_k)$ through future states Z_{k+l} , so $H_{k+1}^{(w)} \epsilon(Z_k)$ is a future-weighted bilinear functional of the trajectory.

Define

$$\mu_k^{(w)} = \mathbb{E}_\pi [H_{k+1}^{(w)} \epsilon(Z_k)].$$

Then

$$S_n - \mathbb{E}S_n = -\alpha \sum_{k=1}^{n-1} \left\{ H_{k+1}^{(w)} \epsilon(Z_k) - \mu_k^{(w)} \right\}.$$

Let $\mathcal{F}_k = \sigma(Z_1, \dots, Z_k)$. The Markov property gives

$$\mathbb{E} [H_{k+1}^{(w)} \epsilon(Z_k) \mid \mathcal{F}_k] = v_k^{(w,\varepsilon)}(Z_k),$$

where

$$v_k^{(w)}(z) = \sum_{l=1}^{n-k} B^{n-k-l+1} (Q^l \tilde{A})(z) B^{l-1}, \quad v_k^{(w,\varepsilon)}(z) = v_k^{(w)}(z) \epsilon(z).$$

Under stationarity, $\mu_k^{(w)} = \pi(v_k^{(w,\varepsilon)})$. Therefore

$$S_n - \mathbb{E}S_n = -\alpha(U_M + U_R),$$

with

$$U_M = \sum_{k=1}^{n-1} \left\{ H_{k+1}^{(w)} \epsilon(Z_k) - v_k^{(w,\varepsilon)}(Z_k) \right\}$$

and

$$U_R = \sum_{k=1}^{n-1} \left\{ v_k^{(w,\varepsilon)}(Z_k) - \pi(v_k^{(w,\varepsilon)}) \right\}.$$

This decomposition splits the centered statistic into two structurally different parts: U_R is an ordinary centered additive functional of the original Markov chain, while U_M is a future-centered bilinear term — the conditional expectation given $\mathcal{F}_k$ vanishes summand-wise, but the summands are not forward martingale differences. We bound each piece separately.

Step 1: bound on U_R . Because $\pi(\tilde{A}) = 0$, applying Q^l followed by integration against π replaces $\tilde{A}$ by its l -step propagation away from stationarity. UGE then gives the geometric Dobrushin bound

$$\|(Q^l \tilde{A})(z)\| = \left\| \int \tilde{A}(u) \{Q^{l(z, du)} - \pi(du)\} \right\| \leq 2C_A \Delta(Q^l) \leq 2C_A (1/4)^{\lfloor l/t_{\text{mix}} \rfloor}.$$

Inserting this into $v_k^{(w)}$ and factoring out the slow rate $(1 - \alpha a)^{(n-k)/2}$ from the two B -powers, the inner sum becomes a fast-decaying l -series:

$$\begin{aligned} \|v_k^{(w)}\|_\infty &\leq C \sum_{l=1}^{n-k} (1 - \alpha a)^{(n-k-l+1)/2} \Delta(Q^l) (1 - \alpha a)^{(l-1)/2} \text{ (triangle + Lyapunov)} \\ &\leq C (1 - \alpha a)^{(n-k)/2} \sum_{l=1}^{\infty} (1/4)^{\lfloor l/t_{\text{mix}} \rfloor} \quad \text{(extend to } l = \infty) \\ &\leq C t_{\text{mix}} (1 - \alpha a)^{(n-k)/2} \quad \text{(geometric block sum).} \end{aligned}$$

Multiplying by $\|\epsilon\|_\infty$ gives

$$\|v_k^{(w, \epsilon)}\|_\infty \leq C t_{\text{mix}} \|\epsilon\|_\infty (1 - \alpha a)^{(n-k)/2}.$$

The functions $v_k^{(w, \epsilon)}(Z_k) - \pi(v_k^{(w, \epsilon)})$ are centered under π and uniformly bounded by the previous display, so the weighted Markov concentration/Rosenthal bound for centered time-dependent functions yields

$$\begin{aligned} \|U_R\|_{L_p} &\leq C p^{1/2} t_{\text{mix}}^{1/2} \left(\sum_{k=1}^{n-1} \|v_k^{(w, \epsilon)}\|_\infty^2 \right)^{1/2} \quad \text{(Rosenthal)} \\ &\leq C p^{1/2} t_{\text{mix}}^{3/2} \|\epsilon\|_\infty \left(\sum_{k=1}^{n-1} (1 - \alpha a)^{n-k} \right)^{1/2} \quad \text{(plug previous bound)} \\ &\leq C p^{1/2} t_{\text{mix}}^{3/2} \|\epsilon\|_\infty \frac{1}{\sqrt{\alpha a}} \quad \text{(geometric series).} \end{aligned}$$

Step 2: bound on U_M . Project onto a deterministic unit vector u and unfold $H_{k+1}^{(w)} u$ as a sum over future states:

$$H_{k+1}^{(w)} u = \sum_{l=1}^{n-k} g_{k,l}(Z_{k+l}), \quad g_{k,l}(z) = B^{n-k-l+1} \tilde{A}(z) B^{l-1} u.$$

Each $g_{k,l}$ is centered under π (since $\pi(\tilde{A}) = 0$), and the two B -powers give the uniform bound

$$\|g_{k,l}\|_\infty \leq C (1 - \alpha a)^{(n-k)/2},$$

which is independent of l . Squaring and summing over l produces only linear growth in $n - k$:

$$\sum_{l=1}^{n-k} \|g_{k,l}\|_\infty^2 \leq C (n - k) (1 - \alpha a)^{n-k}.$$

Note the contrast with the U_R analysis: there, UGE folded the l -sum into a single t_{mix} factor, whereas here the same l -independent bound is applied $(n - k)$ times — the price of conditional centering being weaker than π -centering.

Applying the Markov concentration lemma to the future chain conditionally on $\mathcal{F}_k$,

$$\|H_{k+1}^{(w)}u - \mathbb{E}[H_{k+1}^{(w)}u \mid \mathcal{F}_k]\|_{L_p} \leq Cp^{1/2}t_{\text{mix}}^{1/2}\sqrt{(n-k)(1-\alpha)^{n-k}},$$

and multiplying by $\epsilon(Z_k)$ (which is $\mathcal{F}_k$ -measurable and bounded by $\|\epsilon\|_\infty$) gives

$$\|H_{k+1}^{(w)}\epsilon(Z_k) - v_k^{(w,\epsilon)}(Z_k)\|_{L_p} \leq Cp^{1/2}t_{\text{mix}}^{1/2}\|\epsilon\|_\infty\sqrt{(n-k)(1-\alpha)^{n-k}}.$$

The summands in U_M are centered conditionally on $\mathcal{F}_k$, but they are *not* forward martingale differences: the kernel $H_{k+1}^{(w)}$ peeks into the future of Z_k . To assemble the per- k bounds into a bound on the sum we therefore invoke the weighted Burkholder/block-coupling estimate for future-centered bilinear Markov sums of Samsonov et al. (2025, Proposition 9), specialized to the present constant-stepsizes kernel:

$$\|U_M\|_{L_p} \leq Cp \left(\sum_{k=1}^{n-1} \|H_{k+1}^{(w)}\epsilon(Z_k) - v_k^{(w,\epsilon)}(Z_k)\|_{L_p}^2 \right)^{1/2}.$$

Plugging in the previous bound and evaluating the resulting sum:

$$\begin{aligned} \|U_M\|_{L_p} &\leq Cp^{3/2}t_{\text{mix}}^{1/2}\|\epsilon\|_\infty \left(\sum_{k=1}^{n-1} (n-k)(1-\alpha)^{n-k} \right)^{1/2} \quad (\text{plug previous bound}) \\ &\leq Cp^{3/2}t_{\text{mix}}^{1/2}\|\epsilon\|_\infty \frac{1}{\alpha a} \quad \left(\text{use } \sum_m m x^m \leq Cx^{-2} \right). \end{aligned}$$

The extra factor $1/\sqrt{\alpha a}$ relative to U_R is precisely the cost of using conditional rather than stationary centering.

Step 3: assembly. Combining the two pieces via $S_n - \mathbb{E}S_n = -\alpha(U_M + U_R)$ and the triangle inequality,

$$\begin{aligned} \|S_n - \mathbb{E}S_n\|_{L_p} &\leq \alpha (\|U_M\|_{L_p} + \|U_R\|_{L_p}) \\ &\leq C\|\epsilon\|_\infty \left(p^{3/2}t_{\text{mix}}^{1/2}\frac{1}{a} + p^{1/2}t_{\text{mix}}^{3/2}\sqrt{\frac{\alpha}{a}} \right). \end{aligned}$$

The first term (from U_M) is the leading contribution for small α ; the second (from U_R) carries the heavier t_{mix} dependence but vanishes as $\alpha \rightarrow 0$. Multiplying by α gives the asserted bound for $T_n^{(1,\alpha)} = -\alpha S_n$. $\square$

Remark 2. The stationarity assumption is used only to avoid boundary notation in this local calculation. For a non-stationary initial law, UGE gives the same bulk estimate plus the standard exponentially small mixing boundary terms. The bias of $T_n^{(1,\alpha)}$ is a separate issue and is controlled through the stationary expansion of $\mathbb{E}J_n^{(1,\alpha)}$ in Levin et al. (2025).